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Soil Bulk Density assessments

Bulk density is an important parameter for understanding the physical, chemical and biological soil properties. Dry bulk density and total porosity are indicators to characterize the state of compactness of a topsoil.

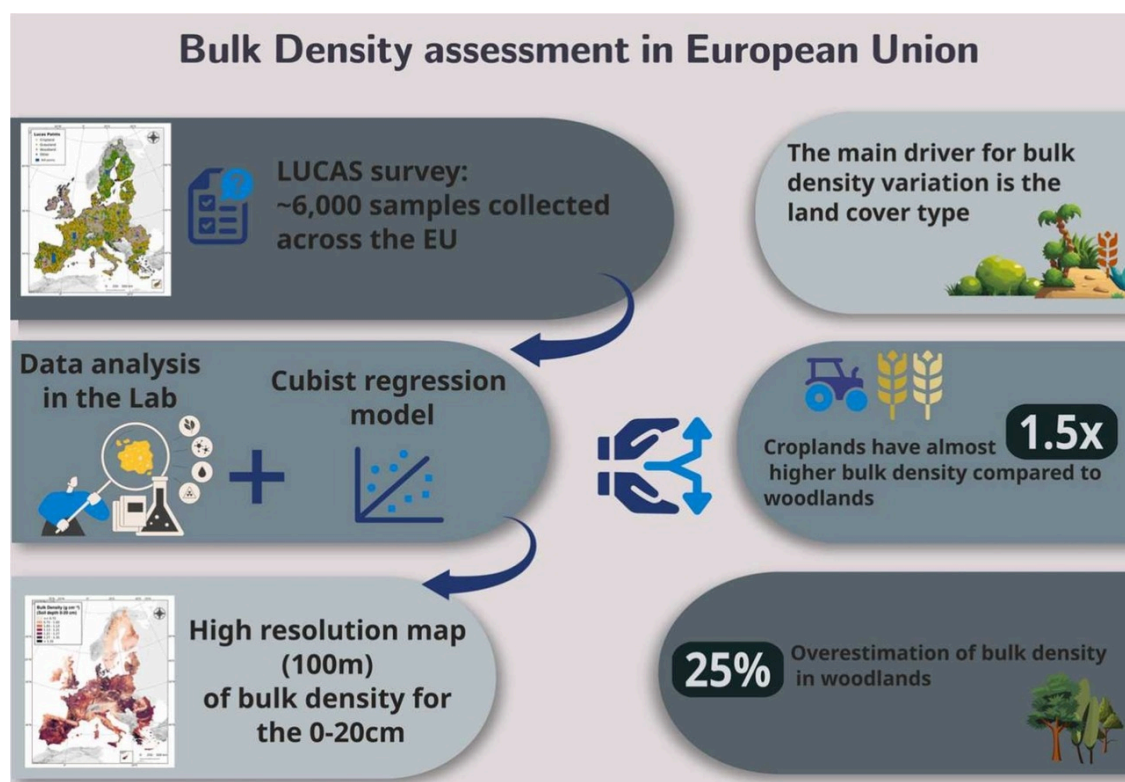
Accurate bulk density data is important for the determination of soil porosity and soil moisture. Bulk density is inversely related to soil porosity which for air and water movement. Bulk density is used as a proxy indicator to determine the soil compaction stress in topsoils. In addition, the bulk density penetration stress and therefore influence the availability of fertilizers to plants and their efficiency. Accurately measuring bulk density is crucial for carbon stocks and their changes in time and space. The precise determination of soil bulk density holds particular significance in carbon crediting credits based on the absolute amount of carbon sequestered. However, it is important to acknowledge that measurements of soil bulk density are prone to error and can reach up to 40%. Moreover, in soils with high rock fragment content (> 30 vol%), this error is further exacerbated, reaching up to 100% if gravel is not accounted. Bulk density is calculated as the ratio of dried soil mass to its volume (Blake, 1965; Hillel, 1980):

$$\rho_b = M_s / V_s$$

ρ_b is estimated as $Mg\ m^{-3}$ while M_s is the weight (Mg) and V_s the volume of the sampled dry soil (in m^3). In many other cases bulk density is reported in $g\ cm^{-3}$.

Data inputs: The LUCAS topsoil survey includes ~20,000 points in the EU with measured physical, chemical, and biological properties. The LUCAS sampling campaign followed the 2009 and 2015. The LUCAS 2018 soil introduced new modules for analysis in a limited number of samples due to budget constraints. The modules include the assessment of soil biodiversity, pesticides residues, assessment of soil erosion features and bulk density. For the 2018 LUCAS campaign, bulk density was measured on approximately 6000 locations across EU+UK. Even if the total LUCAS topsoil survey the bulk density analysis was limited to the 6000 points due to budget constraints. Bulk density was determined at various depth levels: 0–10 cm (61 points), and 20–30 cm (140 points, only for Portugal). Moreover, for locations where measurements were available for both 0–10 cm and 10–20 cm the 0–20 cm stratum was computed by averaging the measurements from the 0–10 cm and 10–20 cm depths across 5659 points.

The spatial interpolation of soil bulk density was performed using the Cubist regression trees.

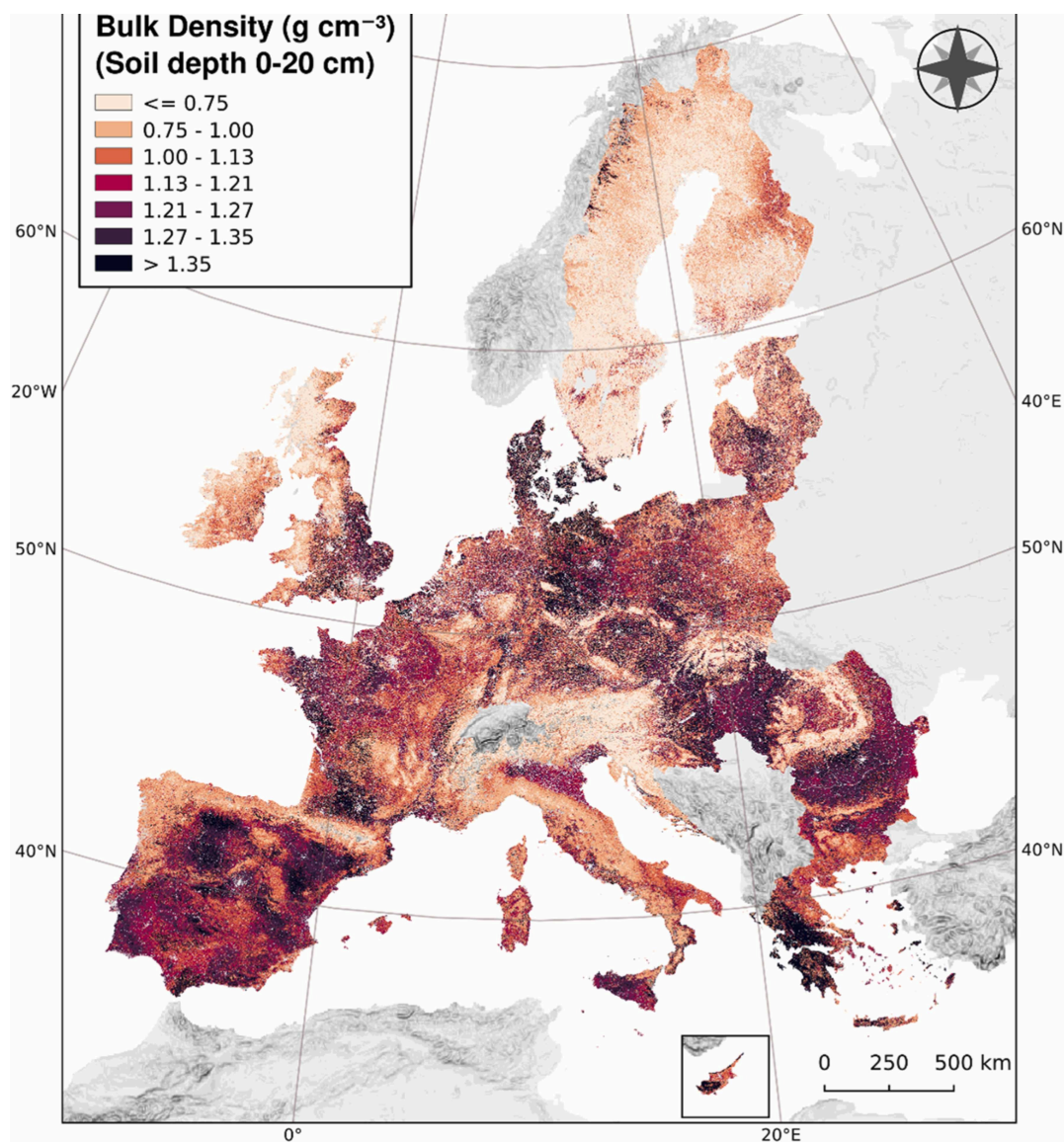


The median bulk density for the 0–10 cm is $1.1\ g\ cm^{-3}$ (mean: $1.04\ g\ cm^{-3}$) with the 25th percentile at $0.85\ g\ cm^{-3}$ and the 75th percentile at $1.29\ g\ cm^{-3}$. For the 10–20 cm is $1.18\ g\ cm^{-3}$ (mean: $1.13\ g\ cm^{-3}$) which is 7% higher compared to the 0–10 cm. This layer has the 25th percentile at $0.95\ g\ cm^{-3}$ and the 75th percentile at $1.41\ g\ cm^{-3}$.

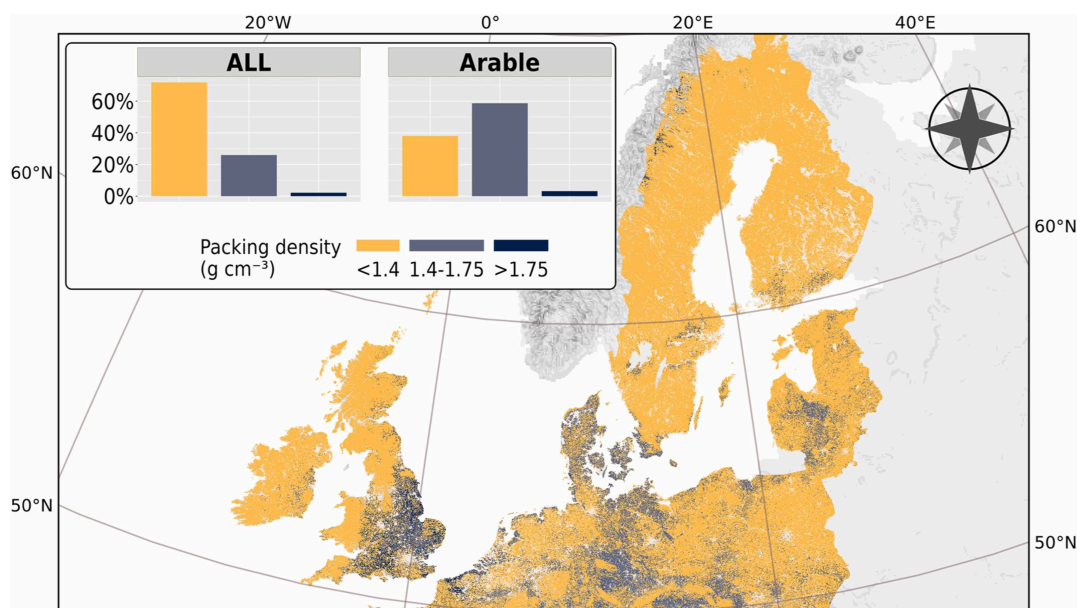
The main driver for bulk density variation is the land cover type. Croplands have almost 1.5 times higher bulk density compared to woodlands (mean BD at $1.26\ g\ cm^{-3}$ followed by permanent crops (BD= $1.23\ g\ cm^{-3}$), heterogeneous agricultural areas (BD= $1.14\ g\ cm^{-3}$), pastures (BD= $1.08\ g\ cm^{-3}$) and woodlands (BD= $0.84\ g\ cm^{-3}$). In most cases, forest soils have much lower values in their upper layer (0–10 cm) due to its richness in organic matter. Compared to past estimates of bulk density which were based on pedotransfer rules, we found an overestimation of bulk density in woodlands in the measured bulk density in LUCAS 2018 survey.

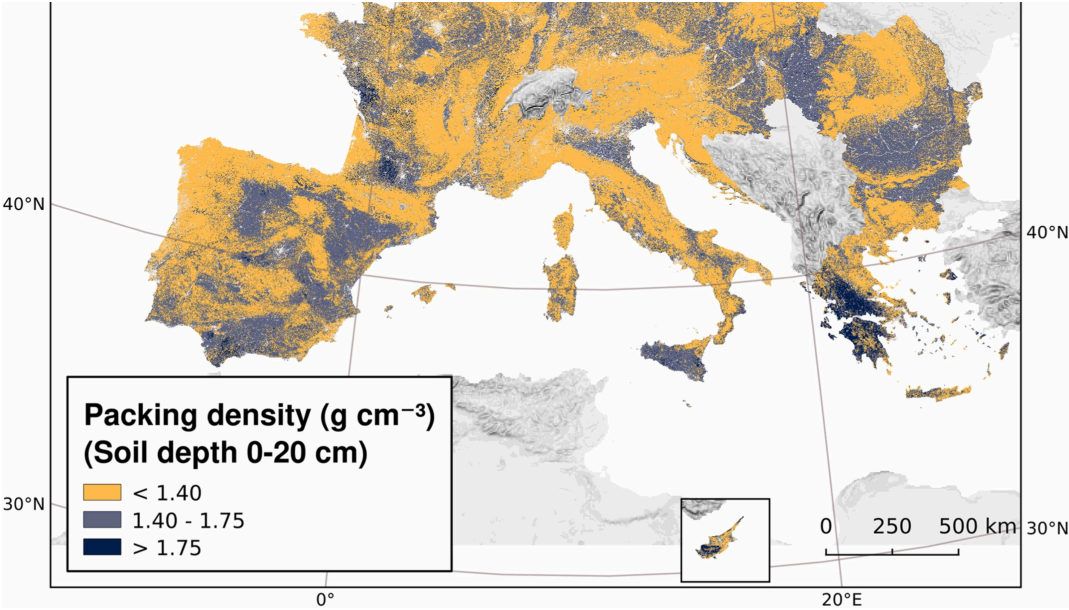
The pan European assessment of bulk density does not challenge any local or regional assessment which have developed with higher density of data. We acknowledge that the LUCAS topsoil database of 6000 samples is limited for a pan European study but this is the most comprehensive EU survey to date. The accessibility of points did not allow to have a much higher number of samples for the analysis. The sampling time is also an important factor which influences bulk density values. The management practices of a farmer (tillage, harvesting) may influence the composition of the sample and the derived BD values. Livestock density and grazing are important missing information for better interpreting bulk density in LUCAS. The size of the coring cylinder (rings) and in-situ soil moisture content significantly affect BD accuracy.





Bulk density is mainly driven by land cover type with croplands having the highest BD and woodlands the lowest in all depths. As the soil BD is the then the relationship between soil compaction and its capacity to store and transport water or air is obvious. As the vertical stress in soils is part of r bulk density can estimate this vertical stress to topsoil. Soil compaction is a major threat to soils particularly in intensively agricultural systems. Soil agricultural productivity, decrease crop yields, decrease water infiltration and accelerate run-off and risk of soil erosion. We estimated the packing de rule and as inputs the bulk density map and the clay content. The Packing Density (PD) is a measure of compactness of the soil and can be a usefu interpretation of the degree of soil compaction. The PD equation ($\text{PD} = \text{BD} + 0.009 \times \text{C}$) was initially developed to estimate soil compaction for the (C has been proposed as a proxy indicator for soil compaction.





Download the data

Reference: Panagos, P., De Rosa, D., Liakos, L., Labouyrie, M., Borrelli, P., Ballabio, C., 2024. Soil bulk density assessment in Europe. *Agriculture*, **364**: 108907. <https://doi.org/10.1016/j.agee.2024.108907>

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Soil Compaction